Multiple Integrals

$\int x^n dx = \frac{x^{n+1}}{n+1}$ $\qquad \int a^x dx = \frac{a^x}{\ln a}$ $\qquad \int \frac{1}{x^2-a^2} dx = \frac{1}{2a} \ln\left(\frac{x-a}{x+a}\right)$

$\int \frac{1}{x} dx = \ln x$ $\qquad \int \frac{1}{x^2+a^2} dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)$ $\qquad \int \frac{1}{a^2-x^2} dx = \frac{1}{2a} \ln\left(\frac{a+x}{a-x}\right)$

$\int \frac{1}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right)$ $\qquad \int \frac{1}{\sqrt{x^2+a^2}} = \log\left|x+\sqrt{x^2+a^2}\right|$ $\qquad \int \frac{1}{\sqrt{x^2-a^2}} dx = \log\left|x+\sqrt{x^2-a^2}\right|$

$\int \frac{1}{\sqrt{x}} = 2\sqrt{x}$

Type 1

$$\int_0^1\int_0^x xy\,dx\,dy$$

$$\int_0^1 x\left(\int_0^x y\,dy\right)dx$$

Integrating limit me $x \Rightarrow y$ को Integrate करो taking Integrating limits same $(0 \to x)$

Type 2

$$\int_0^2\int_0^y xy\,dx\,dy$$

$$\int_0^2 y\left(\int_0^y x\,dx\right)dy$$

Integrating limit me $y \Rightarrow x$ को Integrate करो $(0 \to y)$

Type 3

$$\int_0^2\int_0^1 xy\,dy\,dx$$

↑ Jah pahle likha use kar

$$\cancel{\left(\int_0^2 x\,dx\right)\cdot}\left(\int_0^1 y\,dy\right)$$

y लिखा पहले उसे कर

x वाला limit = dy

y वाला limit = dx

$$\int_0^2\left[x\int_0^1 y\,dy\right]dx$$

$$\int_0^2 x\left(\frac{1}{2}\right)dx$$

$$\frac{1}{2}\left[\frac{x^2}{2}\right]_0^2 \quad \frac{1}{2}\cdot\frac{4}{2} = 1$$

que) $\int_0^2\int_0^{\sqrt{2x}} xy\,dx\,dy$

$$\int_0^2 x\left(\int_0^{\sqrt{2x}} y\,dy\right)dx \Rightarrow \int_0^2 \frac{x}{2}\left[y^2\right]_0^{\sqrt{2x}} dx \Rightarrow \frac{2}{2}\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}$$

que) $\int_0^1\int_0^x (x^2+y^2)x\,dy\,dx$ ***

$$\int_0^1\int_0^x (x^3+xy^2)\,dy\,dx$$

$$\int_0^1\left(\int_0^x x^3 dy + x\int_0^x y^2 dy\right)dx \Rightarrow \int_0^1\left(x^4+\frac{x^4}{3}\right)dx \Rightarrow \int_0^1 \frac{4}{3}x^4 dx$$

$$= \frac{4}{3}\times\frac{1}{5}\times\left[x^5\right]_0^1 = \frac{4}{15}$$

que) $\int_0^1\int_0^{\sqrt{1+x^2}} \frac{1}{1+x^2+y^2}\,dy\,dx$

$$\int_0^1\left(\int_0^{\sqrt{1+x^2}} \frac{1}{(\sqrt{1+x^2})^2+y^2}\,dy\right).dx$$

$$\int_0^1 \left[\left(\frac{1}{\sqrt{1+x^2}}\right)\tan^{-1}\left(\frac{y}{\sqrt{1+x^2}}\right)\right]_0^{\sqrt{1+x^2}} \cdot dx$$

$$\Rightarrow \int_0^1 \frac{1}{\sqrt{1+x^2}}\tan^{-1}\left(\frac{\sqrt{1+x^2}}{\sqrt{1+x^2}}\right)dx$$

$$\frac{\pi}{4}\int_0^1 \frac{1}{\sqrt{1+x^2}}\,dx$$

$$\cancel{\left[\tan^{-1}(x)\right]_0^1}$$

$$\cancel{\frac{\pi^2}{16}} \quad \frac{\pi}{4}\left[\log\left(x+\sqrt{1+x^2}\right)\right]_0^1$$

$$\frac{\pi}{4}\log\left(1+\sqrt{2}\right)$$

Que)

$$\int_0^{a\sqrt{3}} \int_0^{\sqrt{x^2+a^2}} \frac{x}{y^2+x^2+a^2}\, dx\, dy$$

$$\int_0^{a\sqrt{3}} x \left(\int_0^{\sqrt{x^2+a^2}} \frac{1}{y^2+\sqrt{(x^2+a^2)^2}}\, dy \right) dx$$

$$\int_0^{a\sqrt{3}} x \left(\int_0^{\sqrt{x^2+a^2}} \frac{1}{y^2+(\sqrt{x^2+a^2})^2} \right) dx$$

$$\int x \frac{1}{\sqrt{x^2+a^2}} \left[\tan^{-1}\left(\frac{y}{\sqrt{x^2+a^2}} \right) \right]_0^{\sqrt{x^2+a^2}} \cdot dx$$

$$\int_0^{a\sqrt{3}} \frac{x}{\sqrt{x^2+a^2}} \left(\frac{\pi}{4} \right) dx = \frac{\pi}{4}\left(\frac{1}{2}\right) \int_0^{a\sqrt{3}} \frac{2x\, dx}{\sqrt{x^2+a^2}}$$

$x^2+a^2 = t$

$\cancel{2x\,dx = 2t\,dt}$

$\cancel{2x = dt}$

$2x \frac{dx}{dt} = 1$

$2x\,dx = dt$

$x = 0$

$3a^2 + a^2 = t$

$$\frac{\pi}{8} \int_{a^2}^{4a^2} \frac{dt}{\sqrt{t}}$$

$$\Rightarrow \left[2\sqrt{t} \right]_{a^2}^{4a^2} \frac{\pi}{8}$$

$$= 2\left(\sqrt{4a^2} - \sqrt{a^2} \right) \frac{\pi}{8} = \cancel{2}\,(2a-a) \frac{\pi}{\cancel{8}\,4}$$

$$= \frac{\pi a}{4}$$

...the order of Integral

$$\iint (\ldots, y)\,dy\,dx \rightarrow \int_{y=}^{y=}\int_{x=}^{x=} f(x,y)\,dx\,dy$$

...rocess If are given with y limits initially, then change it to x

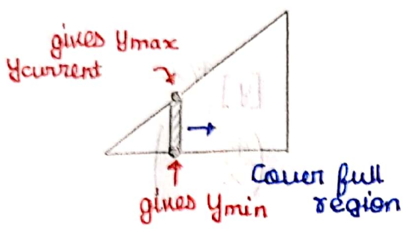
... write given limits $x_1 = \bigcirc$ to $x_2 = \bigcirc$ and $y_1 = \bigcirc$ to $y_2 = \bigcirc$

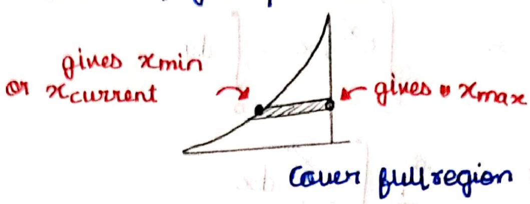
...raw the curves by getting y as a function of x $y = f(x)$

$x = 4y$ आयेगा But you draw $y = x/4$ ✓

③ जिसकी limit निकालनी है, strip उसके parallel मायेगी, x कि निकालनी है, x के parallel, y की निकालनी है। y के parallel

④



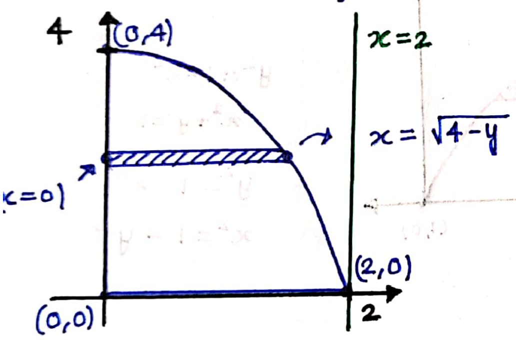


जो constant होगा uska direct min/max करके निकालो।

Q1) $$\int_0^2\int_0^{4-x^2} \frac{x\cdot e^{2y}}{4-y}\,dy\,dx$$

$y_i = 0$ to $y_f = 4-x^2$

$x_i = 0$ to $x_f = 2$



To find : limits of x

$x = 0$ to $x = \sqrt{4-y}$

$y = 0$ to $y = 4$

$$\int_0^4\int_0^{\sqrt{4-y}} \frac{x\cdot e^{2y}}{4-y}\,dx\,dy$$

$$\int_0^4 \frac{e^{2y}}{4-y}\,dy \int_0^{\sqrt{4-y}} x\,dx$$

$$\frac{1}{2}\int_0^4 \frac{e^{2y}}{4-y} \times (4-y)$$

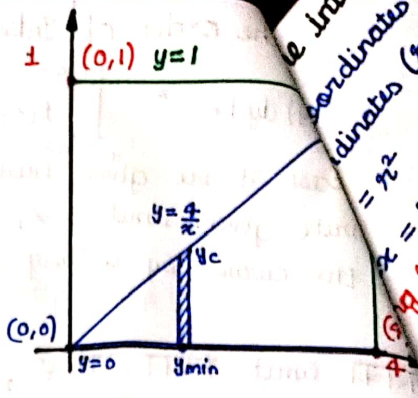
$$= \frac{1}{2}\int_0^4 e^{2y}\,dy = \frac{1}{4}\left[e^{2y}\right]_0^4 = \frac{e^8-1}{4}$$

Q) $\displaystyle\int_0^1 \int_{4y}^{4} e^{x^2}\,dx\,dy$

$x = 4y$ to $x = 4$

$y = 0$ to $y = 1$

To find : limits of y



$y = 0$ to $y = 4/x$

$x = 0$ to $x = 4$

$$\int_0^4 \int_0^{4/x} e^{x^2}\,dx\,dy = \int_0^4 e^{x^2}dx \int_0^{4/x} dy$$

$[y]_0^{4/x}$

$\left(\frac{4}{x} - 0\right)$

$$= \frac{1}{4}\int \frac{x e^{x^2}}{\cancel{x}}\,dx$$

$$= \frac{1}{4}\int_0^4 e^{x^2} x\,dx = \frac{1}{4}\int_0^{16} e^t\,dt$$

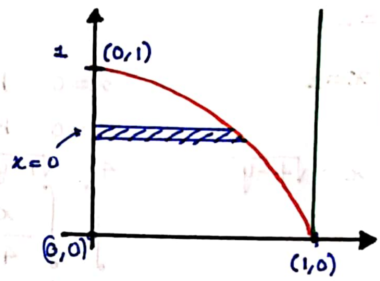
$x^2 = t$

$x = 0 \quad t = 0$

$x = 4 \quad t = 16$

$$= \left(\frac{e^{16}-1}{4}\right)$$

Q) $\displaystyle\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}}\,dx\,dy$



$y^2 = 1 - x^2$

$x^2 + y^2 = 1$

$y^2 = 1 - x^2$

$x^2 = 1 - y^2$

$x = 0$ to $x = 1$

$y = 0$ to $y = \sqrt{1-x^2}$

To find : limits of x

$y = 0$ to $y = 1$

$x = 0$ to $x = \sqrt{1-y^2}$

$$\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{e^y}{(e^y+1)(\sqrt{1-x^2-y^2}}\,dx\,dy$$

$a = \sqrt{1-y^2}$

$$= \int_0^1 \frac{e^y}{e^y+1}\,dy \int_0^{\sqrt{1-y^2}} \frac{1}{\sqrt{(\sqrt{1-y^2})^2 - x^2}}\,dx$$

$\dfrac{1}{\sqrt{a^2-x^2}}$

$$\left[\sin^{-1}\left(\frac{x}{\sqrt{1-y^2}}\right)\right]_0^y = \sin^{-1}(1) - \sin^{-1}(0)$$

$$= \frac{\pi}{2} - 0$$

$$= \frac{\pi}{2}\int_0^1 \frac{e^y\,dy}{e^y+1}$$

$e^y = t$

$y = 0 \quad t = 1$

$y = 1 \quad t = e$

$$\frac{\pi}{2}\int_1^e \frac{dt}{t+1} = [\ln(t+1)]_1^e$$

$$= \frac{\pi}{2}\ln\frac{(e+1)}{2}$$

...he integral from Cartesian to polar coordinates

...oordinates (x,y)

...dinates (r,θ)

$...= r^2$, $x = r\cos\theta$, $y = r\sin\theta$, $\tan\theta = y/x$

$...x = r\,dr\,d\theta$

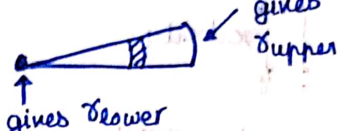
...ng limits of r and θ , r always starts from 0 and goes till a certain point

* r ke liye ज्यादा दिमाग मत लगा , जिसपे strip slide कर रही है uski equation me r substitute कर के r की value निकाल लो ||

* θ के लिये दिमाग लगा सकते हैं ||

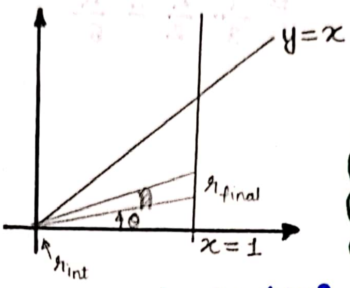
Algorithm :-

1. Basics you already know
2. Now see strips would look like



3. Just find out eqⁿ of curve where r up lies , change it to polar eqⁿ and get r

$$\int_0^1\int_0^x (x+y)\,dy\,dx$$



$y=0$ to $y=x$

$x=0$ to $x=1$

r_{final} slides on line $x=1$

$x = r\cos\theta$

$r\cos\theta = 1 \Rightarrow r = \sec\theta$, $r=0$ to $r=\sec\theta$

θ goes from 0 to $\pi/4$

$$\int_0^{\pi/4}\int_0^{\sec\theta} r^2(\cos\theta+\sin\theta)\,dr\,d\theta = \int_0^{\pi/4}(\cos\theta+\sin\theta)\,d\theta \int_0^{\sec\theta} r^2\,dr$$

$$= \int_0^{\pi/4}\left[\left(\frac{\cos\theta}{\cos^3\theta}\right)+\left(\frac{\sin\theta}{\cos^3\theta}\right)\right]d\theta = \int_0^{\pi/4}\sec^2\theta\,d\theta + \int_0^{\pi/4}\tan\theta\sec^2\theta\,d\theta$$

$\tan\theta = t$

$\sec^2\theta\,d\theta = dt$

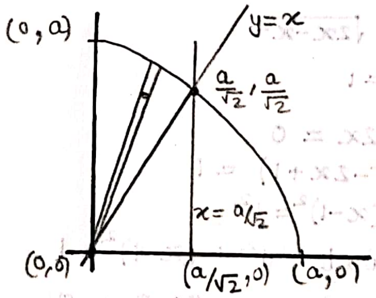
$\theta = 0 \quad t = 0$

$\theta = \pi/4 \quad t = 1$

$$= [\tan\theta]_0^{\pi/4} + \left[\frac{t^2}{2}\right]_0^1$$

$$= 1 + \frac{1}{2} = \left(\frac{3}{2}\right)\frac{1}{3} = \frac{1}{2}$$

Q) $$\int_0^{a/\sqrt{2}}\int_x^{\sqrt{a^2-x^2}} \frac{x\,dy\,dx}{\sqrt{x^2+y^2}}$$



$x^2+y^2=a^2$

$\frac{a^2}{2}+y^2=a^2$

$y^2=\frac{a^2}{2}$

$y=0$ to $y=\sqrt{a^2-x^2}$

$x=0$ to $x=\frac{a}{\sqrt{2}}$

strip lies on circle $x^2+y^2=a^2$

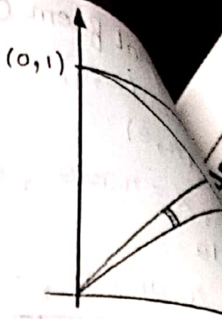
$r^2=a^2 \Rightarrow r=a$

r goes from 0 to a

θ from $\pi/4$ to $\pi/2$

$$\int_{\pi/4}^{\pi/2}\int_0^a \frac{r\cos\theta\,r\,dr\,d\theta}{r} = \int_{\pi/4}^{\pi/2}\cos\theta\,d\theta \int_0^a r\,dr = \frac{a^2}{2}\left(1-\frac{1}{\sqrt{2}}\right)$$

Que) $\displaystyle\int_0^1\int_0^{\sqrt{1-y^2}}(x^2+y^2)\,dy\,dx$



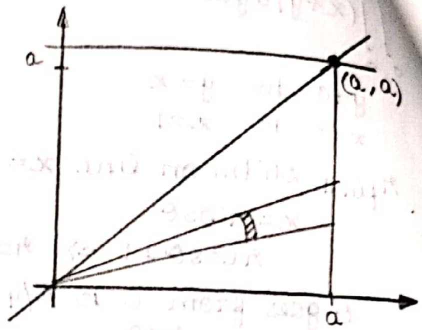
$x=0$ to $x=\sqrt{1-y^2}$
$y=0$ to $y=1$

r lies on $x^2+y^2=1$
r goes from $r=0$ / $r=1$ $\quad\theta$ from 0 to $\pi/2$

$$\int_0^{\frac{\pi}{2}}\int_0^1 r^2\cdot r\,dr\,d\theta = \int_0^{\frac{\pi}{2}}d\theta\int_0^1 r^2\,dr$$

$$= \frac{1}{3}\cdot\frac{\pi}{2} = \frac{\pi}{6}$$

Que) $\displaystyle\int_0^a\int_y^a x\,dx\,dy$



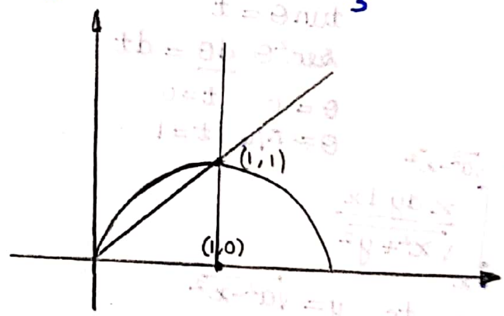
$x=y$ to $x=a$
$y=0$ to $y=a$

r lies on $x=a$ $\qquad r\cos\theta = a \Rightarrow r = a\sec\theta$
r goes from $r=0$ to $r=a\sec\theta$
θ from 0 to $\pi/4$

$$\int_0^{\frac{\pi}{4}}\int_0^{a\sec\theta} r\cos\theta\cdot r\,dr\,d\theta = \int_0^{\frac{\pi}{4}}\cos\theta\,d\theta\int_0^{a\sec\theta} r^2\,dr$$

$$= \int_0^{\frac{\pi}{4}}\cos\theta\times\frac{a^3}{3\cos^3\theta}\,d\theta = \frac{a^3}{3}\left[\tan\theta\right]_0^{\pi/4} = \frac{a^3}{3}$$

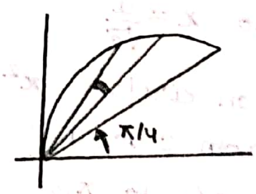
★ Que) ★ $\displaystyle\int_0^1\int_x^{\sqrt{2x-x^2}}(x^2+y^2)\,dy\,dx$



$y=x$ to $y=\sqrt{2x-x^2}$
$x=0$ to $x=1$

$$y^2+x^2-2x=0$$
$$y^2+(x^2-2x+1)=1$$
$$y^2+(x-1)^2=1^2$$

r slides on circle $y^2+(x-1)^2=1$



$$r^2\sin^2\theta + r^2\cos^2\theta - 2r\cos\theta = 0$$
$$r^{\cancel{2}} = 2\cancel{r}\cos\theta \qquad \theta \text{ from } \pi/4 \text{ to } \pi/2$$

$r=0$ to $r=2\cos\theta$

$$\int_{\frac{\pi}{4}}^{\pi/2}\int_0^{2\cos\theta} r^2\cdot r\,dr\,d\theta = \int_{\pi/4}^{\frac{\pi}{2}}d\theta\int_0^{2\cos\theta} r^3\,dr = \int_{\pi/4}^{\pi/2}\frac{2^4\cos^4\theta}{4}\,d\theta$$

$\cos 2\theta = 2\cos^2\theta - 1$
$\dfrac{1+\cos 2\theta}{2}$

$$= 4\int_{\pi/4}^{\pi/2}(\cos^2\theta)^2\,d\theta = \int_{\pi/4}^{\pi/2}(1+\cos 2\theta)^2\,d\theta$$

$$\frac{\Gamma\frac{p+1}{2}\,\Gamma\frac{q+1}{2}}{2\,\Gamma}$$

$$= \int\left(1+2\cos 2\theta+(\cos 2\theta)^2\right)d\theta = [\theta]_{\pi/4}^{\pi/2} + \frac{2}{2}[\sin 2\theta]_{\pi/4}^{\pi/2} + \left[\frac{1+\cos 4\theta}{2}\right]_{\pi/4}^{\pi/2}$$

...culate करने वाले सवाल

$\int$ dxdy , ये इन सवालो का Template है जहा 3 – 4 Curve / eqⁿ दे देंगे और उनके Beech का enclosed area निकालना होगा

R → Represents region

...in Task Here is to get limits for Integration :-

...Draw the figure first : once you have a closed figure you're all set

2) Now Limits : we can take strips of y or x , कोई सा भी ले सकते है

But → agar question मे एक strip से difficult पड रहा है तो switch the strip on spot!

⟨ y की strip से y का limit मिलेगा
x की strip से x का limit मिलेगा ⟩

3) Now Integrate, once you have limits.

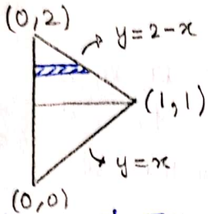
g the order of Integral : Integral changes to 2 Integrals

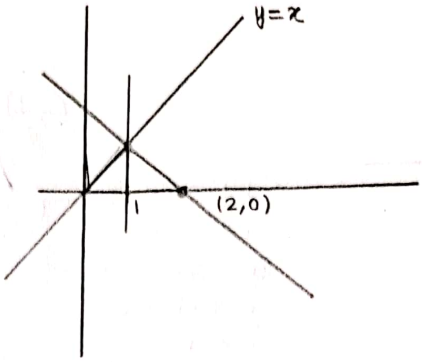
नहीं इसमें strips को तोड़ कर Integrate करना है simple Idea

$$\int\int^{x} \left(\frac{x}{y}\right) dy\,dx$$

$y = x$ to $y = 2-x$

$x = 0$ to $x = 1$





Limit given : of ~~x~~ y, to find : of x

$$\int_0^1\int_0^y \left(\frac{x}{y}\right) dx\,dy + \int_1^2\int_0^{2-y} \left(\frac{x}{y}\right) dx\,dy$$

$$\int_0^1 \frac{1}{y} \times \frac{y^2}{2}\,dy + \int_1^2 \frac{1}{y}\,\frac{(2-y)^2}{2}\,dy = \int_0^1 \frac{y}{2}\,dy + \frac{1}{2}\int\left(\frac{4}{y} + \frac{y^2}{y} - \frac{2y\cdot 2}{y}\right) dy$$

$$= \frac{1}{2}\times\frac{1}{2}(1) + \frac{1}{2}\left[4\ln y + \frac{y^2}{2} - 4y\right]_0^1$$

$$= \frac{1}{4} + \frac{1}{2}\left[\frac{1}{2} - 4\right.$$

② $\displaystyle\int_0^5\int_{2-x}^{2+x} dy\,dx = 25$ (PT)

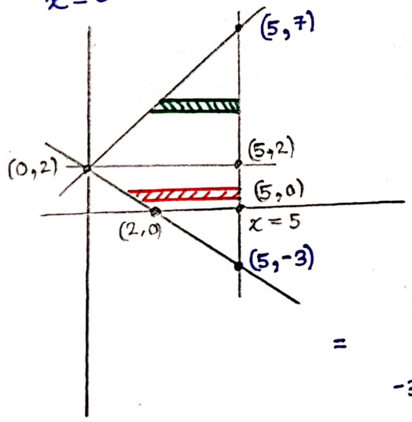
$y = 2-x$ to $y = 2+x$

$x = 0$ to $x = 5$

$$\int_0^5\int_{2-x}^{2+x} dx\,dy$$

want : x limit

$x = 2-y$



$$\int_{-3}^{2}\int_{2-y}^{5} dx\,dy + \int_{2}^{7}\int_{y-2}^{5} dx\,dy$$

$$\int_{-3}^{2} [5-(2-y)]\,dy + \int_{2}^{7} [5-(y-2)]\,dy$$

$$= \int_{-3}^{2} (3+y)\,dy + \int_{2}^{5} (7-y)\,dy$$

$$= \left[3y + \frac{y^2}{2}\right]_{-3}^{2} + \left[7y - \frac{y^2}{2}\right]_{2}^{7}$$

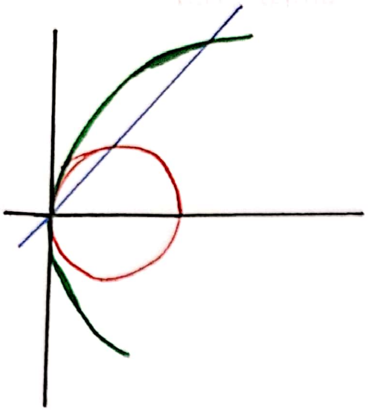
$$= \left[(6+2) - \left(-9+\frac{9}{2}\right)\right] + \left[\left(35 - \frac{25}{2}\right) - (14-2)\right]$$

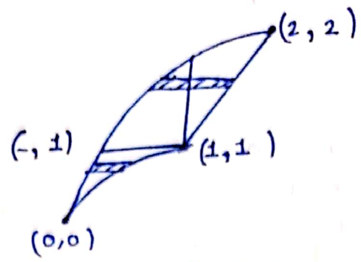
$$= \frac{16}{2} - \left(-\frac{9}{2}\right) + \frac{45}{2} - \frac{24}{2}$$

$$= \frac{(45+16+9)-24}{2} = \frac{45+25-24}{2} = 23$$

6 ~~7~~ 6 10
25
45

$\iint_R (xy)\,dx\,dy$, over the Region given by $x^2+y^2=2x$, $y^2=2x$ and $y=x$





$y^2 = 2y$

$y(y-2)=0$

$x^2+y^2=2x$

$x^2+x^2=2x$

$2x^2-2x=0$

$x(x-1)=0$

$x=1$

x की limit

$x = y^2/2$ से $\quad x=x$

काम

y की तो

$$\int_0^1\int_{\sqrt{2x-x^2}}^{\sqrt{2x}} (xy)\,dy\,dx + \int_1^2\int_x^{\sqrt{2x}} (xy)\,dy\,dx$$

$$\frac{1}{2}\int_0^1 x\left(2x-(2x-x^2)\right)dx + \frac{1}{2}\int_1^2 x(2x-x^2)$$

$$\frac{1}{2}\int_0^1 x(x^2)\,dx + \frac{1}{2}\int_1^2 (2x^2-x^3)\,dx$$

$$\frac{1}{2}\int_0^1 x^3\,dx + \frac{1}{2}\left(\int_1^2 2x^2\,dx - \int_1^2 x^3\,dx\right)$$

$$\frac{1}{2}\left[\frac{1}{4} + \frac{2}{3}[2^3-1^3] - \frac{1}{4}[2^4-1^4]\right)$$

$$\frac{1}{2}\left[\frac{1}{4} + \frac{4\times2\times7}{3\times4} - \frac{1}{4}(15)\right]$$

$$\frac{1}{2}\left[\frac{(56+1)-15}{4}\right] = \frac{42}{2\times2\times2} = \frac{21}{4}$$